

Code 1

```
#include <stdio.h>

int main()
{
    int arr[3][4], i, j, k, max;

    for (i = 0; i < 3; i++)
    {
        for (j = 0; j < 4; j++)
        {
            printf("Enter value ");
            scanf("%d", &arr[i][j]);
        }
    }

    max = arr[0][0];

    for (i = 0; i < 3; i++)
    {
        for (int j = 0; j < 4; j++)
        {
            if (max < arr[i][j])
            {
                max = arr[i][j];
            }
        }
    }

    printf("%d is max value\n", max);

    return 0;
```

```
}
```

Code 2

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int arr[2][2], i, j, transpose;
```

```
    for (i = 0; i < 2; i++)
```

```
    {
```

```
        for (j = 0; j < 2; j++)
```

```
        {
```

```
            printf("Enter values of matrix elements\n");
```

```
            scanf("%d", &arr[i][j]);
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
    for (i = 0; i < 2; i++)
```

```
    {
```

```
        for (j = 0; j < 2; j++)
```

```
        {
```

```
            printf("%d \t ", arr[j][i]);
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
    return 0;
```

```
}
```

Code 3

```
#include<stdio.h>

int main(){

    int array[3][3]={{1,4,3},{5,3,2},{4,6,8}};

    int sum=0;


    for(int i=0; i<3; i++){
        for(int j=0; j<3; j++)
        {
            printf("%d",array[i][j]);

        }
        printf("\n");
    }


    for(int i=0; i<3; i++)
    {
        for(int j=0; j<3; j++)
        {
            if(i==0 && j==i || i==1 && j==i || i==2 && j==i)
                sum +=array[i][j];
        }
        printf("\n");
    }

    printf("sum is: %d",sum);
```

```
return 0;
}
```

Code 4

```
#include<stdio.h>

int small_value(int[]);

int main()
{
    int result;

    int array[10] = {1,2,3,4,5,6,7,9,0,11};

    printf("Value if: %d",result = small_value(array));
}

int small_value(int array[])
{
    int count =1;
    for(int i=0; i<10; i++)
        for(int j=0; j<i; j++)
        {
            if(array[j] == count)
                count++;
        }
    return count;
}
```